

IR REMOTE CONTROLS

Ray of Hope for Cerebral Palsy Patients

Imagine not being able to control your body movements. This is the blaze cerebral palsy patients battle with day-in and day-out. CePal, an infrared-based remote control developed by Dhirubhai Ambani Institute of Information and Communication Technology, can now reduce their woes

■ VANISHA JOSEPH

Cerebral palsy is a disorder affecting one's ability to move and maintain posture. Recent estimates reveal that India has more than four million cases of cerebral palsy, including over 200 children born affected every day.

Numerous assistive technology devices, ranging from electronic communication boards with pictures representing an individual's daily needs to electronic speech synthesisers, have been designed to help individuals communicate easily and perform daily tasks. However, most devices need to interface to a computer that is yet to penetrate every Indian household, leaving communication to perform day-to-day tasks an everyday battle for most patients.

CePal, an infrared-based remote control developed by Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar, and now backed by National Trust (an autonomous organisation of the Ministry of Social Justice and Empowerment in India), has finally won this losing battle for cerebral palsy patients.

This lightweight, low-cost device worn on the wrist, hand or head allows people affected with cerebral palsy to operate remote-controlled electronic devices like TV, air-conditioner, lights and fans.

"We took advantage of the fact that most equipment today come with IR remote control. We used the IR receiver built into the device and directly



CePal, worn on the wrist, hand or head, allows people affected with cerebral palsy to operate remote-controlled electronic devices

operated it from 'CePal.' Thereafter, we directly generated the IR control signal needed by the device receiver to operate them; avoiding any intermediate device to act as a gateway. CePal by itself detects and differentiates the optimal tilt, thereby controlling the device. The device allows patients to switch on/off lights and fans, change the TV channel or volume, etc by movement," says Prabhat Ranjan, professor, embedded systems and sensor networks, DA-IICT.

CePal uses a 2D accelerometer sensor (MMA6270QT) to detect motion. The detected movement is then sent to the analogue-to-digital conversion (ADC) pin of an Atmel ATmega32 microcontroller chip. Thereafter, the microcontroller maps digital values from the ADC to IR control signals. The IR signals generated depend upon the IR protocol of the

target device, the direction of the tilt and the action to be performed based on the requirement of the patient.

A cut above the rest!

Started as a project by two students, Ramya and Pallavi, who wanted to work on assistive technology for the physically challenged, CePal has taken its first step in becoming the hope for cerebral palsy patients at Sparsh Pediatric Neuro Developmental Therapy Clinic in Ahmedabad. It is being used on cerebral palsy patients and being constantly improved based on the feedback.

But, what makes CePal score over other assistive technologies? It is a complete solution, giving it an edge over its competitors. "Because of their complexity and huge prices, most existing devices are out of reach of the